DEPARTMENT OF MINING ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY(ISM), DHANBAD

NUMERICAL MODELLING PRACTICAL EXAMINATION

Sl. No.:	**Adm. No.:**	**Name:**	Max. Time: 45 Mins
Max. Marks: 50			
Note: Create a folder on Desktop,	Like : ME-(Adm No)		

QUESTION

Aim: Perform FE analysis of a 20m wide rib pillars at a depth of 300m between two horizontal panels of 100m wide each. Study effect of height of the pillar on its strength for height of extraction: 3m for the property set -1.

Property Set 1:

Pillar rock	Roof and floor rock
Uniaxial compressive strength, Co = 5 MPa Coefficient of friction, Φ = 30° **(Coulomb failure criteria)** Modulus of elasticity, E = 2 GPa Poisson' ratio = 0.15 Unit Weight, ρ = 1400 kg/m³	Uniaxial compressive strength, Co = 10 MPa Coefficient of friction Φ = 35° Modulus of elasticity, E = 10 GPa Poisson' ratio = 0.25 Unit Weight, ρ = 2200 kg/m³

In-situ Stress: gravity load: (9.8 m/sec² acceleration)

Observation table:

Table 1: Pillar height = 3m

Distance from Centre of the pillar	σ_1 (1)	σ_2 or σ_3 (2)	Strength (3)	Safety Factor (Strength /σ_1)
0m				
2m				
4m				
6m				
8m				
10m				
		Weighted SF		

Mohr-Coulomb shear failure criterion:

Strength = Co + qσ_3; Safety Factor = strength/σ_1

where,

$$q = \tan^2\left(\frac{\pi}{4} + \frac{\phi}{2}\right)$$

σ_1 and σ_3 are major and minor principal stresses in the pillar.

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NUMERICAL MODELLING PRACTICAL EXAMINATION

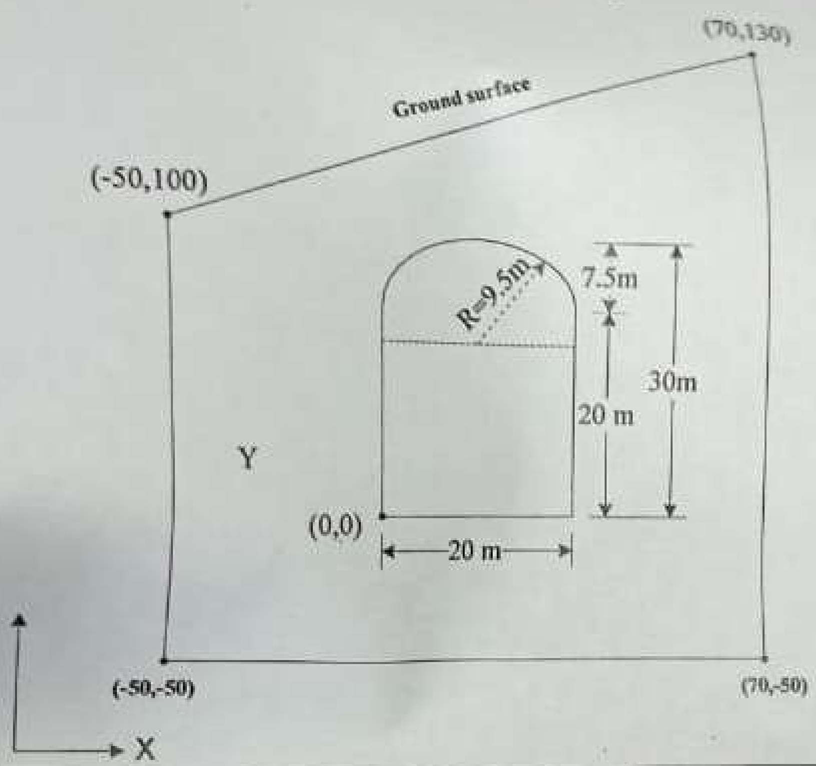
Sl. No.:		
Max. Marks: 50	**Adm. No.:**	
Note: Create a folder on Desktop, Like : MP-(Adm No)		**Name:**
		Max. Time: [illegible]

QUESTION

Aim: Perform Finite Element Analysis of Stresses around of an underground excavation in massive rockmass and observe the changes in stress distribution around the excavation.

Rock propertires are: density = 2650 kg/m^3, Young's modulus = 50 GPa, poisson's ratio = 0.22.

Insitu horizontal stress along X-axis is 2 times the vertical stress.



Observation Table

Table 1: σ_{xx} & σ_{yy} at the nodes along x-axis (size of the elements at the boundary = ...)

Distance from center of the excavation, m	Insitu results (1)		FEA results (2)		Difference (1 – 2)		Remarks
	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	

Table 2: σ_{xx} & σ_{yy} at the nodes along y-axis (size of the elements at the boundary = ...)

Distance from center of the excavation, m	Insitu results (1)		FEA results (2)		Difference (1 – 2)		Remarks
	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	

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NUMERICAL MODELLING PRACTICAL EXAMINATION

Sl. No.: [illegible]	Adm. No.: [illegible]	Name: [illegible]
Min. Marks: 30		
Note: Create a folder in Desktop,	Like: MI_Adm No.	Max. Time: 45 Mins

Aim: Perform FE analysis of a 20m wide rib pillars at a depth of 300m between two horizontal panels of 100m wide each. Study effect of height of the pillar on its strength for height of extraction 5m, for the property set -1.

Property Set 1:

Pillar rock		Roof and floor rock	
Uniaxial compressive strength, C_0	= 5 MPa	Uniaxial compressive strength, C_0	= 10 MPa
Coefficient of friction, Φ (Coulomb failure criteria)	= 30°	Coefficient of friction Φ	= 35°
Modulus of elasticity, E	= 2 GPa	Modulus of elasticity, E	= 10 GPa
Poisson' ratio	= 0.15	Poisson' ratio	= 0.25
Unit Weight, ρ	= 1400 kg/m^3	Unit Weight, ρ	= 2200 kg/m^3

In-situ Stress: gravity load (9.8 m/sec^2 acceleration)

Observation table:

Table 2: Pillar height = 5m

Distance from Centre of the pillar	σ_1 (1)	σ_2 or σ_3 (2)	Strength (3)	Safety Factor (Strength/σ_1)
0m				
2m	[illegible]	[illegible]	[illegible]	[illegible]
4m	[illegible]	[illegible]	[illegible]	[illegible]
6m	[illegible]	[illegible]	[illegible]	[illegible]
8m	[illegible]	[illegible]	[illegible]	[illegible]
10m	[illegible]	[illegible]	[illegible]	[illegible]
		Weighted SF	= [illegible]	

Mohr-Coulomb shear failure criterion

Strength = $C_0 + q\sigma_3$; Safety Factor = strength/σ_1

where,

$$q = \tan^2\left(\frac{\pi}{4} + \frac{\phi}{2}\right)$$

σ_1 and σ_3 are major and minor principal stresses in the pillar.

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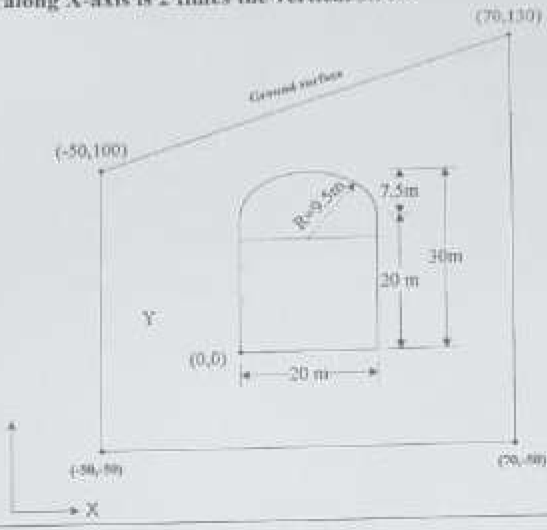
SL No.: 4	Adm. No.: [illegible]	Name: [illegible]
Max. Marks: 50		
Note: Create a folder on Desktop,	Like : ME-(Adm No)	

QUESTION

Aim: Perform Finite Element Analysis of Stresses around of an underground excavation in massive rockmass and observe the changes in stress distribution around the excavation.

Rock propertires are: density = 2650 kg/m³, Young's modulus = 50 GPa, poisson's ratio = 0.22.

Insitu horizontal stress along X-axis is 2 times the vertical stress.



Observation Table

Table 1: σ_{xx} & σ_{yy} at the nodes along x-axis (size of the elements at the boundary = ...)

Distance from center of the excavation, m	Insitu results (1)		FEA results (2)		Difference (1 – 2)		Remarks
	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	
	-1.4095	-2.795	-0.1491	-3.801	-1.2605	1.0061	
	-1.3424	-2.6636	-1.2051	-3.940	-0.1375	1.2701	
	-1.3201	-2.541	-2.30	-0.5021	0.9491	-2.0091	

Table 2: σ_{xx} & σ_{yy} at the nodes along y-axis (size of the elements at the boundary = ...)

Distance from center of the excavation, m	Insitu results (1)		FEA results (2)		Difference (1 – 2)		Remarks
	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	
	-1.3121	-2.618	-2.304	-2.0123	0.9044	-0.6061	
	-1.8447	-2.769					

QUESTION

Aim: A cross section through a oil storage cavern is shown in Figure 1. Cross sectional dimnsions and excavation sequences of the storage cavern, construction tunnel and water tunnels are shown in Fig 1.

Rock propertires are: density = 2650 kg/m^3, Young's modulus = 50 GPa, poisson's ratio = 0.22, UCS = 3.5 MPa and angle of internal friction of rock is 40°.

H = 24

Insitu horizontal stress along x-axis is 2 times the vertical stress.

Using FEM stress analysis of the oil storage cavern, perform its stabilty analysis and support requirement of the oil storage tunnel.

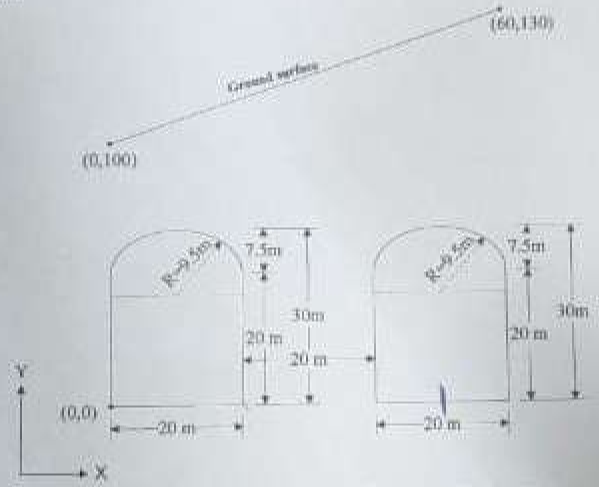


Figure -1: X-section through oil storage caverns

Observation Table:

Table 1: σ_{xx} & σ_{yy} at the nodes along x-axis (size of the elements at the boundary = ...)

Distance from center of the cavern, m	Insitu results (1)		FEA results (2)		Difference (1 – 2)		Remarks
	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	σ_{xx}	σ_{yy}	
1	15.174	11.319	14.78	11.218	0.36	0.003	
3	13.76	11.21	13.53	11.1	0.23	0.11	
5	12.13	9.55	11.87	9.2	0.26	0.15	
7	11.17	9.137	10.97	9.01	0.20	0.126	